

Moving Towards New Frontiers with Generative AI

Generative AI is making its mark in diverse fields, but it must be nurtured with care to ensure it's free of bias and preserves human values.



while Transformers originally designed for sequence tasks have become central to large scale language and multimodal generation.

The rise of large foundation models, which can be adapted to many tasks after pre-training, has further unified progress in the field. Prior surveys have compared technical trade-offs and discussed ethical implications such as deepfakes and bias. Building on this literature, our work provides an updated overview that connects recent technical advances with practical applications and ongoing challenges.

Technical foundations of generative AI

Generative AI is built on machine learning models that learn the structure of data and create new samples from it. The four major model families behind modern systems are Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), diffusion models, and Transformer-based models. We briefly describe the intuition behind each approach and highlight their main strengths and limitations. Figure 1 presents a GAN architecture, while Table 1 compares these paradigms.

Generative Adversarial Networks (GANs): Generative Adversarial Networks (GANs) model learning as a competition between two neural networks: a generator and a discriminator. Proposed by Goodfellow *et al*, the generator G produces synthetic samples from random noise, while the discriminator D attempts to distinguish real data from generated outputs. Both

Generative AI is a branch of artificial intelligence focused on producing new data that resembles its training examples. Unlike discriminative models that classify or detect patterns, generative models create original outputs such as text, images, music, or video. By learning the structure of data, these systems can produce realistic and novel content, making them influential across creative industries, research, and automation.

Recent advances in model scale and performance have accelerated interest in generative AI. Large language models can draft articles or hold conversations, image systems generate visuals from text prompts, and audio models synthesize speech or music. This progress marks a shift in AI's role from analytical assistance to creative collaboration.

However, these capabilities also introduce risks. Deepfakes, factual inaccuracies, and biased outputs highlight the need for responsible development and oversight. Hence, understanding generative AI requires

both technical knowledge and ethical awareness.

The journey so far

Research on generative models spans several decades in machine learning. Early work relied on probabilistic approaches such as Gaussian mixture models and hidden Markov models, which established the basic principles of modelling data distributions. The deep learning era introduced more powerful frameworks, most notably Generative Adversarial Networks (GANs) by Goodfellow *et al* and Variational Autoencoders (VAEs) by Kingma and Welling, which renewed widespread interest in generative modelling (see References at the end of this article).

Subsequent developments included flow-based models with exact likelihood training and autoregressive approaches like PixelCNN and WaveNet that generated data sequentially. In recent years, diffusion models and Transformer based architectures have achieved major breakthroughs. Diffusion methods now lead in image and audio synthesis,